

Problem 1

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Problem 1 (a) Suppose $f'(2)$ exists. Which of the following must be true?

- (i) $\lim_{x \rightarrow 2} f(x)$ must exist, but $\lim_{x \rightarrow 2} f(x) \neq f(2)$
- (ii) $\lim_{x \rightarrow 2} f(x) = f(2)$.
- (iii) $\lim_{x \rightarrow 2} f(x) = f'(2)$
- (iv) $\lim_{x \rightarrow 2} f(x)$ need not exist.

Explanation. The correct answer is (ii):

$$\begin{aligned} f'(2) \text{ exists} &\iff f \text{ differentiable at } x = 2 \\ &\implies f \text{ continuous at } x = 2 \\ &\iff \lim_{x \rightarrow 2} f(x) = f(2) \end{aligned}$$

Assuming that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$, which of the following is true?

- (b) (i) $\frac{0}{0} = 1$
- (ii) the tangent line to $y = \sin(x)$ at $(0, 0)$ has slope 1.
- (iii) you can cancel the x 's.
- (iv) for all x near 0, $\sin(x) = x$.
- (v) for all x near 0, $\sin(x) \approx x$.

Explanation. This problem has two correct answers: (ii) and (v).

Statement (ii) is true:

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{\sin(0+h) - \sin(0)}{h} \text{ Which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \\ &= 1 \\ &\implies \text{slope of tangent line at } (0, 0) \text{ is } 1 \end{aligned}$$

Statement (v) is true: When x is near 0 the tangent line $y = x$ is a good approximation to f .